

Please scan the QR code to check in this session

Mastering shared Windows devices in your organization

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Solution architect



Security
Windows and Device



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What are we going to discuss?



Architecture and core concepts



Basic configuration options



Additional configurations that should be considered



Deployment options and assignment considerations

Concepts

Architecture and core concepts

Characteristics of shared devices

What are the main characteristics of shared devices?



Many different
users



Same experience
every time*



No personal device
ownership

Important decisions to make

What are the important decisions to make before starting?

Managing expectation of the device

What configuration model should be used for the shared devices?



Shared PC

Multiple people sign in on the device, one at a time



Kiosk

Device is locked to a dedicated app or task

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Handling identity of the user

What user account should be used for the shared devices?



Entra ID account Local account



Conditional Access and MFA



Local storage of account information

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Handling assignments to device

What would be the best assignments for shared devices?



No primary user

Be careful with applications and scripts that follow the user



Filters

Use filters, when possible, to target specific devices only

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Thinking about device licensing

What licensing should be used for shared devices?

Often covered by existing licenses of users using the device. Especially when talking about Shared PCs



Might be required when device is used by users without eligible licenses, like: Temporary workers, local or shared accounts, public access

Device licenses do not support multi-app protection profiles, Conditional access, and user-based management features

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Managing expectation of the device

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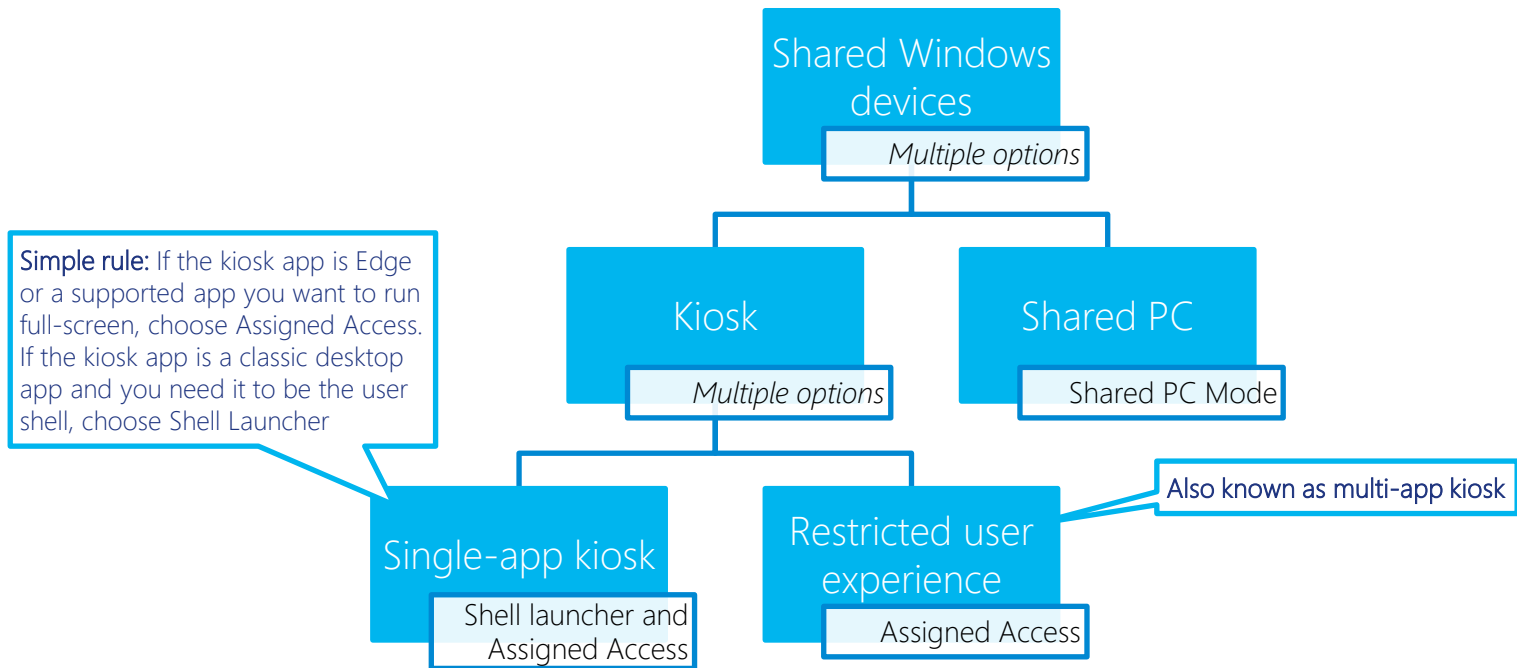
Device licenses do not support Intune app protection policies, Conditional Access, and user-based management features.

Configuration

Basic configuration options

Choosing the deployment model

What are the technical options for configuring shared devices?



Shared PC Mode basics

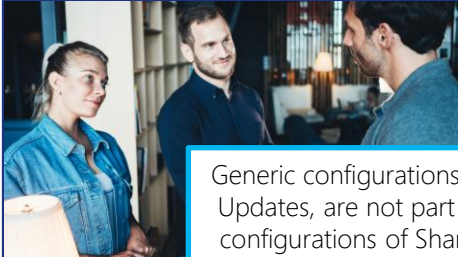
What are the main configurations that are part of Shared PCs?



Consider using Personal Data Encryption in combination with OneDrive

With or without OneDrive

Exclusions are possible
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\SharedPC\Exemptions\



Generic configurations, like Windows Updates, are not part of the default configurations of Shared PC Mode.

Account management

- Account manager
- Account model



Advanced customizations

- Power
- Sleep
- Sign-in
- Maintenance
- Pagefile
- Local storage

Technical reference: [Shared PC technical reference | Microsoft Learn](#)

What does it look like in Intune?

The screenshot displays the Microsoft Intune admin center interface. The top navigation bar shows 'Microsoft Intune admin' and 'Microsoft Intune admin center'. The left-hand navigation pane lists various configuration categories, with 'Shared multi-user' selected. The central configuration area, titled 'Create profile', shows a list of settings for a 'Shared multi-user' profile. The right-hand pane, titled 'Settings picker', shows a search bar and a list of settings categorized by 'Shared PC'.

Microsoft Intune admin center

Home > Devices | Configuration

Shared multi-user
Windows 10 and later

Basics **Config**

Configure access, account, and Windows 10 or later used

In organizations where managed account manager settings

[Click here for more info.](#)

Shared PC mode

Guest account

Account management

Local Storage

Power Policies

Sleep time out (in seconds)

Previous Next

Create profile ...
Windows 10 and later - Settings catalog

Sleep Timeout 300

Inactive Threshold 30

Enable Shared PC Mode With One Drive Sync Not configured

Enable Shared PC Mode false

Enable Account Manager false

Disk Level Deletion 25

Disk Level Caching 50

Deletion Policy Delete at disk space threshold

Account Model Guest

Previous Next

Settings picker
Use commas "," among search terms to lookup settings by their keywords

Search for a setting Search

+ Add filter

Browse by category

- Secure Boot
- Security
- Settings
- Shared PC**
- Smart Screen
- Speech
- Start
- Storage

17 settings in "Shared PC" category [Select all these settings](#)

Setting name

- ☒ Account Model
- ☒ Deletion Policy
- ☒ Disk Level Caching
- ☒ Disk Level Deletion
- ☒ Enable Account Manager
- ☒ Enable Shared PC Mode
- ☒ Enable Shared PC Mode With One Drive Sync
- ☒ Inactive Threshold

Assigned Access basics

Generic configurations, like Windows Updates and Power, are not part of the default configurations.

Profile

Allowed apps

Start layout

Taskbar

Account

Autologon

Technical reference: [Assigned Access Policy Settings | Microsoft Learn](#)



What does it look like in Intune?

Microsoft Intune admin center

Home > Devices | Configuration

Kiosk ...
Windows 10 and later

Basics 2 Configuration settings Assignments Applicability Rules Review + create

Configure your devices 1 Mobile Apps blade, App about Windows kiosk m

Select a kiosk mode *

Custom ...
Windows 10 and later

Basics Configuration settings Assignments Applicability Rules Review + create

OMA-URI Settings ⓘ

Name ↑↓	Description ↑↓	OMA-URI ↑↓	Value
No settings			

Add Export

Add Row

OMA-URI Settings

Name * Multi-app kiosk mode configuration ✓

Description Kiosk mode configuration using Assigned Ac... ✓

OMA-URI * ./Vendor/MSFT/AssignedAccess/Configuration ✓

Data type * String ✓

Value *
<Profile Id="{9A2A490F-10F6-4764-974A-53B19E722C23}">
<AllAppsList>
<AllowedApps>
<App
AppUserModelId="Microsoft.WindowsCalc
ulator_8wekyb3d8bbwe!App" />

Save Cancel

Assigned Access XML basics

```
<?xml version="1.0" encoding="utf-8" ?>
```

```
<AssignedAccessConfiguration xmlns="http://schemas.microsoft.com/AssignedAccess/2017/config">
```

```
<Profiles>
```

```
<Profile Id="{GUID}">
```

Each profile defines a set of applications that are allowed to run

```
<!-- Add configuration here as needed -->
```

```
</Profile>
```

```
</Profiles>
```

```
<Configs>
```

```
<Config>
```

Each config associates a user account or a group to a profile

```
<!-- Add configuration here as needed -->
```

```
</Config>
```

```
</Configs>
```

```
</AssignedAccessConfiguration>
```

Each version is used to determine which features are available for the configuration

Assigned Access versioning

	Version	Alias	Namespace
StartPins	Windows 11, version 22H2	v5	http://schemas.microsoft.com/AssignedAccess/2022/config
TaskbarLayout	Windows 11, version 21H2	v4	http://schemas.microsoft.com/AssignedAccess/2021/config
ClassicAppArguments	Windows 10	v5	http://schemas.microsoft.com/AssignedAccess/202010/config
ClassicAppPath	Windows 10	v3	http://schemas.microsoft.com/AssignedAccess/2020/config
AllowRemovableDrives	Windows 10	rs5	http://schemas.microsoft.com/AssignedAccess/201810/config
GlobalProfile	Windows 10	default	http://schemas.microsoft.com/AssignedAccess/2017/config
AutoLaunch			
AutoLaunchArguments			

Assigned Access XML profiles

<Taskbar ShowTaskbar="true"/>

Each app specifies an app that is allowed through AppLocker in the kiosk experience

Each app specifies an app that is pinned to the startmenu in the kiosk experience

Shows the taskbar in the kiosk experience (can also be used to pin apps)

Assigned Access XML config

```
<Configs>
  <Config>
    <AutoLogonAccount rs5:DisplayName="Kiosk user"/>
    <DefaultProfile Id="{GUID}"/>
  </Config>
</Configs>
```

Uses an autologon account that will be automatically created.

```
<Configs>
  <v3:GlobalProfile Id="{GUID}"/>
</Configs>
```

Uses the same configuration for every user that signs in.

```
<Config>
  <Account>AzureAD\user@contoso.onmicrosoft.com</Account>
  <DefaultProfile Id="{GUID}"/>
</Config>
```

Uses a specific configuration for the specified user or group.

Can be used for local user (.user), Entra user (example), AD user (domain\user), and groups.

What does it look like in Intune?

Microsoft Intune admin center

Home > Devices | Configuration

Custom ...

Windows 10 and later

1 Basics 2 Configuration settings 3 Assignments 4 Applicability Rules 5 Review + create

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Add Export

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Previous Next

Add Row

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<AutoLogonAccount
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<DefaultProfile Id="{9A2A490F-10F6-
4764-974A-53B19E722C23}"/>
</Config>
</Configs>
</AssignedAccessConfiguration>

Save Cancel

Recommendations

Additional configurations that should be considered



Recommendations

Tuning Windows
Updates

Removing power
settings

Disabling keyboard
shortcuts

Deploying apps

Preventing
notifications

Troubleshooting
and logs

Technical reference: [Assigned Access Recommendations | Microsoft Learn](#)



Tune Windows updates

At least for kiosk devices and depending on the shared device usage

Microsoft Intune admin center

Home > Devices | Windows updates

Create Update ring for Windows 10 and later

Windows 10 and later

Enable pre-release builds * ☐ Enable ☒ Not Configured

Select pre-release channel: Windows Insider - Release Preview

User experience settings

Automatic update behavior ☐ Auto install and restart at maintenance time

Active hours start * ☐ 8 AM

Active hours end * ☐ 5 PM

Option to pause Windows updates ☐ Enable ☒ Disable

Option to check for Windows updates ☐ Enable ☒ Disable

Change notification update level ☐ Turn off all notifications, including restart warnings

Use deadline settings ☐ Allow ☒ Not configured

Deadline for feature updates ☐ Number of days, 0 to 30

Deadline for quality updates ☐ Number of days, 0 to 30

Grace period ☐ Number of days, 0 to 7

Auto reboot before deadline ☐ Yes ☒ No

Previous Next

Auto install outside active hours or alternatively configure scheduled install times.

Turn off notifications

Remove power settings

At least for kiosk devices and depending on the shared device usage

Shared device configuration contains default power settings.

Microsoft Intune admin center

Home > Devices | Configuration

Create profile ...

Windows 10 and later - Settings catalog

Administrative Templates

System > Power Management > Sleep Settings

19 of 20 settings in this subcategory are not configured

Specify the system sleep timeout (plugged in) Disabled

System > Power Management > Video and Display Settings

5 of 6 settings in this subcategory are not configured

Turn off the display (plugged in) Disabled

Start Menu and Taskbar

102 of 103 settings in this subcategory are not configured

Remove and prevent access to the Shut Down, Restart, Sleep, and Hibernate Disabled

Previous Next

Settings picker

Use commas "," among search terms to lookup settings by their keywords

Search for a setting

+ Add filter

Browse by category

- Personal Data Encryption
- Personalization
- PKCS certificate
- PKCS imported certificate
- Power
- Printer Provisioning
- Printers
- Privacy

14 settings in "Power" category

Select all these settings

Setting name

- ☐ Select Lid Close Action Plugged In
- ☐ Select Power Button Action On Battery
- ☒ Select Power Button Action Plugged In
- ☐ Select Sleep Button Action On Battery
- ☒ Select Sleep Button Action Plugged In
- ☐ Turn Off Hybrid Sleep On Battery
- ☐ Turn Off Hybrid Sleep Plugged In
- ☐ Unattended Sleep Timeout On Battery

Prevent the device from falling asleep and/or being turned off.

Mostly Power and Power Management related settings.

Use Local Policies Security Options to remove permissions for shutdown.

Disable keyboard shortcuts

At least for kiosk devices and depending on the shared device usage

The Keyboard Filter feature can be used to suppress specific unwanted key combinations, and it works with physical keyboards, the Windows on-screen keyboard, and the touch keyboard.

Enable the Windows feature.

```
Enable-WindowsOptionalFeature -Online -FeatureName Client-KeyboardFilter -All -NoRestart
```

```
$CommonParams = @{"namespace"="root\standardcimv2\embedded"}
$CommonParams += $PSBoundParameters
function Enable-Predefined-Key($Id) {
    $predefined = Get-WMIObject -class WEKF_PredefinedKey @CommonParams |
        where {
            $_.Id -eq "$Id"
        };
    if ($predefined) {
        $predefined.Enabled = 1;
        $predefined.Put() | Out-Null;
        Write-Host Enabled $Id
    }
    else {
        Write-Error "$Id is not a valid predefined key"
    }
}
Enable-Predefined-Key "Alt+Tab"
```

Disable the specified key combinations.



Deploying apps

At least for kiosk devices and depending on the shared device usage

Apps that follow the user leave a trace of installations.

1

Deploy apps to devices

E.g. OneDrive on shared devices, or Explorer in kiosk mode.

2

Do not use an app that might expose (personal) data

3

App must be installed before the configuration

Apps that are later installed cannot be used in configuration.

4

Microsoft Edge has its own kiosk mode

Provides automatic limited experience.

5

App cannot launch another app in kiosk mode

Breaks automatic behavior of kiosk.

6

Prevent apps that require user sign-in in kiosk mode

Kiosk mode is locked down with AppLocker.

Prevent lockscreen notifications

At least for kiosk devices and depending on the shared device usage

Microsoft Intune admin center

Home > Devices | Configuration

Create profile

Windows 10 and later - Settings catalog

1 Basics 2 **Configuration settings** 3 Scope tags 4 Assignments 5 Review + create

+ Add settings ⓘ

^ Above Lock Remove category

1 of 2 settings in this category are not configured

Allow Toasts ⓘ Block ⓘ

^ Administrative Templates Remove category

System > Logon Remove subcategory

23 of 24 settings in this subcategory are not configured

Turn off app notifications on the lock screen ⓘ Enabled ⓘ

Previous Next

Troubleshooting and log locations

Shared PC

Log location

- C:\Windows\SharedPCSetup.log

Registry location

- HKLM\Software\Microsoft\Windows\CurrentVersion\SharedPC

Assigned Access

Log location

- Applications and Services
Logs > Microsoft > Windows > AssignedAccess > Operational

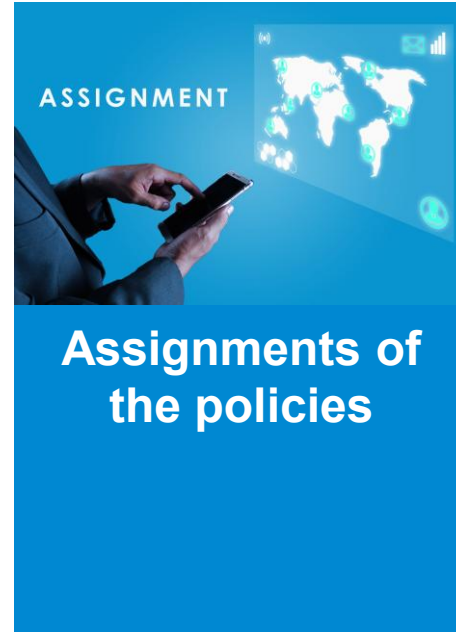
Registry location

- HKLM\Software\Microsoft\Windows\AssignedAccessConfiguration
- HKLM\Software\Microsoft\Windows\AssignedAccessCsp
- HKCU\Software\Microsoft\Windows\AssignedAccessConfiguration

Deployment

Deployment options and assignment considerations

Deployment alignment



Self-deploying devices

Group devices using
(device.devicePhysicalIds
-any (_ -eq
"[OrderID]:KIOSK"))

- * No user interaction required for devices with Ethernet connection
- * Joins the device to Entra ID and enrolls the device into Intune
- * Provisions all policies, applications, certificates, and networking profiles on the device
- * Relies on Enrollment Status Page to prevent access to the device
- * No automatic primary user configuration for the device
- * Requires TPM 2.0

Group/Filter devices
(device.enrollmentPr
ofileName -eq
"Autopilot profile for
Windows kiosk
devices")

Handling assignments

Not easy to combine multiple filters

- 1 Always assign apps and configurations to devices
- 2 Use filters to exclude shared devices from user assignments
- 3 Deploy apps before applying configurations that require the apps
- 4 Don't forget app specific configuration for shared devices

Think about the Microsoft 365 apps activation configuration for shared devices

Layering configurations

The scenario specific configuration that creates the main experience of the device.

Shared PC configuration

Assigned Access configuration

The application layer that contains scenario specific apps.

App

App

App

App

App

App

The recommended configuration layer that contains scenario specific configurations.

Recommended configurations

That includes at least the Windows updates, power settings, keyboard shortcuts, and notifications

The generic security baseline that is applicable to all devices.

Generic security baseline

That includes at least the Windows security baseline



Bonus: Few common challenges

- Autologon is not working on a Window kiosk device
 - Device lock configuration (often part of security baseline)
 - Preferred Entra domain configuration
 - Password policy as part of a compliance policy
- OneDrive configuration is not working on shared Windows device
 - Conflicting configurations
 - Shared PC Mode was already enabled
- Missing settings
 - `./Device/Vendor/MSFT/Policy/Config/Start/HideRecommendedSection`
 - `./User/Vendor/MSFT/Policy/Config/Notifications/DisableAccountNotifications`



Please evaluate this session in the App.

THANK YOU

Are there any questions?





Next session 14:00 – 14:50

**AI-powered infra: From detection to auto-fix
with Azure Copilot**

